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Lights Out! Energy Use in an Average American Home

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The national conversation has turned to solar panels, personal energy generation and Powerwall batteries. Americans have a renewed interest in Kilowatt Hours (kWh), but do we really understand what that means? How many kWhs do you use in an hour? And what in your home uses the most energy? If you are looking to go off-the-grid, you'll need to consider your daily energy usage. Let's break down the average American home based on energy consumption.

First, let's discuss the big numbers. According to the U.S. Energy Information Administration, the average American household uses 10,908 kWh per year, averaging 909 kWh per month. "Louisiana had the highest annual consumption at 15,270 kWh, and Hawaii had the lowest at 6,176 kWh," says (http://1.usa.gov/1GZUxS8) the EIA.

While this is great as an overall figure, when deciding on solar panels you may need more granular information. If you're deciding on how many Tesla Powerwalls you may need to go fully off-grid, you'll need to see what you're using hourly. After all, annual amounts likely won't give you an understanding of how you need to change your habits to achieve day-to-day energy independence.

Remember when reading the values associated with any one appliance or electronic provided below that a kWh refers to an hour's use of that device. Let's take a look at some of the most common household energy culprits. Note that the following values are typical and not indicative of any one brand or model of device. Values provided by Mr. Electricity himself (http://bit.ly/1GQ3KKM), Michael Bluejay.

Kitchen

Those that use an electric oven are in for something of a shock, while not the worst offender in your home, its 2 kWh energy consumption may make you squirm at the prospect of living off grid and making that evening roast. That one hour's use of your electric oven would take a fifth of your entire charge from the largest 10 kWh Tesla Powerwall.

Other larger energy culprits in the kitchen include the microwave oven (1.4 kWh), dishwasher (1.2 kWh) and a coffee maker (0.9 kWh). Pretty much anything that generates heat to cook or clean will be a large energy drain on your reserves.

While hot water is used throughout the house, it is especially useful in the kitchen and your water heater will cost 3.8 kWh for every hour in use. This means you may want to hold off on doing dishes or bathing at night.

Bedroom

Hopefully you keep your bedroom free of electronics, which means you likely only need to worry about lights and a ceiling fan. A typical ceiling fan clocks in at 0.1 kWh, which sounds manageable compared to the energy cost of central air (3.5 kWh).

Living Room

Shockingly, that massive LCD TV of yours really doesn't use that much energy! A 56" flat screen TV only consumes 0.3 kWh, which is only slightly more than a PlayStation 3 (0.2 kWh).

Lighting, which admittedly is a home wide consideration, can be manageable as long as you don't have every light on at once. A single 60 watt bulb only uses 0.1 kWh and a condensed florescent bulb that emits the same amount of light is even less (0.05 kWh). If you are conscious about turning lights off when you leave the room, the energy cost is nearly negligible, but if you forget and leave lights on all over the house, that energy expense will quickly skyrocket.

Heating/Cooling

We already mentioned central air's costly 3.5 kWh, but heating can cost even more! Let's say you're living in a cold climate and you have a 2000 square foot house. Heating that space will cost a shocking 26 kWh! This specific metric is one that may make you consider how many battery packs you'd need to feasibly go off-grid.

Equations

The Energy Department provides (http://1.usa.gov/1G4X4Vg) a simplistic formula for anyone that wants to crunch the numbers themselves on their own energy use per item.

(Wattage × Hours Used Per Day) ÷ 1000 = Daily Kilowatt-hour (kWh) consumption

With this equation and a little bookkeeping on your personal usage, you can figure out the rough numbers on where your energy is going. This also will help when you inevitably decide to curtail certain activities in the hopes of minimizing your energy usage.

Takeaway

We are approaching a time where the average citizen can consider going off the grid. With recent advancements in solar panel conversion and battery storage technology, we are closer than ever to an energy independent future for the average homeowner!

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